MILITARY SPECIFICATION

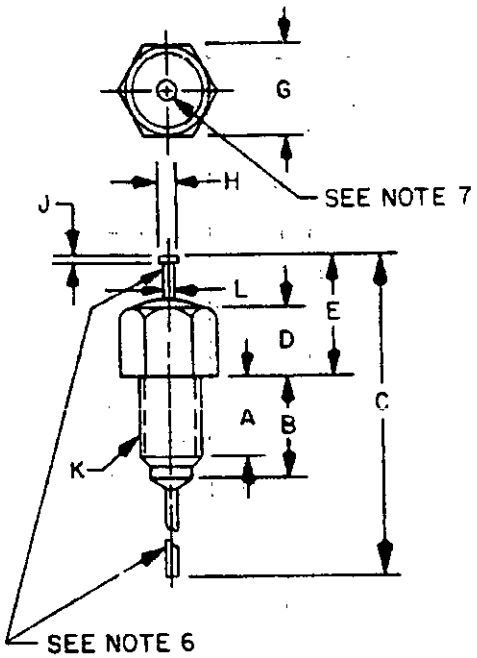
FILTERS, RADIO FREQUENCY INTERFERENCE,

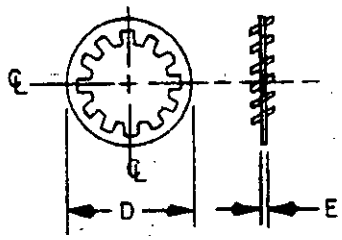
STYLE FL7U

Part numbers M15733/43-0001 and M15733/43-0002 are inactive for new design after the date of this specification. For new design use M15733/61-0002 for M15733/43-0001 and M15733/61-0008 for M15733/43-0002.

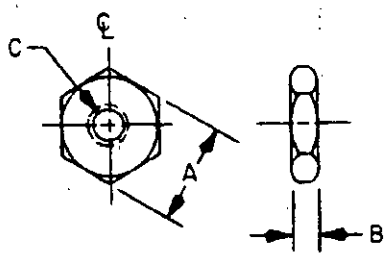
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.

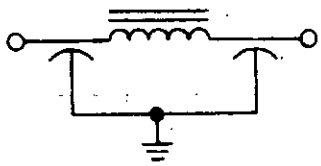




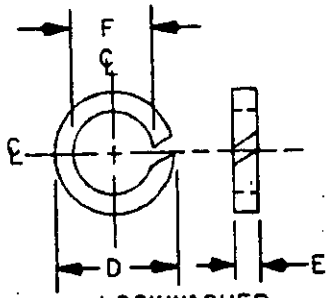
LOCKWASHER
(DASH NO.-0003)



HEX NUT



TYPICAL CIRCUIT DIAGRAM



LOCKWASHER
(DASH NO.-0001 AND-0002)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. Recommended mounting torque: 3 IN-LB (-0001) and 6 IN-LB (-0002 and -0003) maximum.
5. Mounting hardware (lockwasher and hex nut) will be supplied with each filter.
6. Leads shall be silver coated or solder coated copper.
7. Turret head is optional.

FIGURE 1. Case and circuit diagram.

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Filter dimensions.

Dash no.	Dimensions									
	A		B		C		D		E	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
0001	.208 (5.28)	.228 (5.79)	.271 (6.88)	.291 (7.39)	.781 (19.84)	.843 (21.41)	.177 (4.50)	.197 (5.00)	.349 (8.86)	.369 (9.37)
0002	.206 (5.23)	.226 (5.74)	.256 (6.50)	.276 (7.01)	.859 (21.82)	.921 (23.39)	.083 (2.11)	.103 (2.62)	.281 (7.14)	.343 (8.71)
0003	.203 (5.16)	.233 (5.92)	.266 (6.76)	.296 (7.52)	.750 (19.05)	.874 (22.20)	.172 (4.37)	.202 (5.13)	.328 (8.33)	.390 (9.91)

Dash no.	Dimensions								
	G		H		J		K	L	
	Min	Max	Min	Max	Min	Max	THD	Min	Max
0001	.240 (6.10)	.260 (6.60)	.050 (1.27)	.070 (1.78)	.010 (.25)	.030 (.76)	.216-28 UNF-2A	.027 (.69)	.037 (.94)
0002	.177 (4.50)	.197 (5.00)	.050 (1.27)	.070 (1.78)	.010 (.25)	.030 (.76)	.164-32 UNC-2A	.027 (.69)	.037 (.94)
0003	.235 (5.97)	.265 (6.73)	.050 (1.27)	.070 (1.78)	.010 (.25)	.030 (.76)	.216-32 UNEF-2A	.035 (.89)	.045 (1.14)

Hardware dimensions.

Dash no.	Hex nut					Lockwasher			
	A		B		C	D		E	
	Min	Max	Min	Max	Thread	Min	Max	Min	Max
0001	.240 (6.10)	.260 (6.60)	.070 (1.78)	.080 (2.03)	.216-28 UNF-2B	.276 (7.01)	.280 (7.11)	.038 (.97)	.058 (1.47)
0002	.240 (6.10)	.260 (6.60)	.073 (1.85)	.078 (1.98)	.164-32 UNC-2B	.226 (5.74)	.230 (5.84)	.033 (.84)	.053 (1.35)
0003	.302 (7.67)	.322 (8.18)	.070 (1.78)	.080 (2.03)	.216-32 UNEF-2B	.371 (9.42)	.381 (9.68)	.020 (.51)	.024 (.61)

FIGURE 1. Case and circuit diagram - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Metal.

Weight: 2.6 grams, maximum.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 10 amperes maximum.

Voltage drop: 0.10 volts maximum.

Voltage conditioning (quality conformance inspection only). The test shall be performed as follows:

Test temperature: 125°C.

Test voltage: 200 percent of the rated dc voltage. Charging current shall not exceed 50 mA.

Current load: Not applicable.

Points of application of test voltage: Between either terminal and case.

Duration of exposure to test voltage: 100 ±4 hours.

Creepage and clearance: Not applicable.

Seal: Not applicable.

Capacitance to ground: Method 305, MIL-STD-202.

Test frequency: 1,000 Hz ±100 Hz.

Test voltage: 0.5 Vrms at 25°C.

Measured capacitance shall be as specified in table I.

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202. The test voltage shall be 3 times the rated dc voltage applied for 1 to 5 seconds. Charging current shall not exceed 50 mA.

Barometric pressure (reduced): Method 105, test condition E, MIL-STD-202.

Dielectric withstanding voltage: In accordance with initial requirements, except that the test voltage shall be 120 percent of the rated dc voltage applied for 1 to 5 seconds.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Insulation resistance shall be measured at 25°C between either terminal and case, and shall be not less than 10,000 megohms.

Electrification voltage: 100 V dc after 2 minutes (maximum) electrification.

Insertion loss: See table I.

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TABLE I. Electrical characteristics.

Dash number	Rated voltage		Capacitance pF minimum	Minimum insertion loss (dB) in accordance with MIL-STD-220 2/											
				+25°C						-55°C, +125°C					
	DC volts	AC 1/ rms volts		10 MHz	30 MHz	100 MHz	200 MHz	300 MHz	1-10 GHz	10 MHz	30 MHz	100 MHz	200 MHz	300 MHz	1-10 GHz
0001	200	140	1500	---	---	---	45	45	45	---	---	---	28	45	45
0002	100	70	1500	---	---	---	50	50	50	---	---	---	28	50	50
0003	500	---	2500	12	20	45	---	60	70	4	11	25	---	50	70

1/ DC to 400 Hz.

2/ Full-load insertion loss measurements shall be performed at 10 MHz. Measurements above this frequency shall be performed at no-load.

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Overload: In accordance with MIL-F-15733.

Measurements at 25°C after test:

Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition A, MIL-STD-202.

Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion:

Thermal shock: Method 107 test condition B, MIL-STD-202, except that the temperature of step 3 shall be 125°C.

Immersion: Method 104, test condition A, MIL-STD-202.

Measurements after final cycle: Dielectric withstanding voltage with 3 times the rated V dc applied for 1 to 5 seconds, charging current shall not exceed 50 mA.

Insulation resistance shall be not less than 1,000 megohms.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): Method 213, test condition C, MIL-STD-202.

Electrical load: 100 percent of rated voltage between either terminal and case. After completion of the shock test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Vibration, high frequency: Method 204, test condition C, MIL-STD-202, except that part 2 shall be 20 G. Duration shall be 3 hours each in two mutually perpendicular planes.

Electrical load: Rated voltage and rated current shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

Moisture resistance: Method 107, test condition B (except temperature of step 3 shall be 125°C) MIL-STD-202, except that no measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.

Initial measurements: Not applicable.

Polarization voltage: 100 V dc.

Measurements within 24-hour drying period:

Dielectric withstanding voltage shall meet initial requirements.

Insulation resistance shall be not less than 1,000 megohms.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

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Life: Method 108, test condition D, MIL-STD-202.

Test temperature 125°C.

Test voltage: 2 times rated dc voltage with no load at 125°C for 1,000 hours.

Measurements after test:

Dielectric withstanding voltage initial voltage applied for 1 to 5 seconds and shall meet initial requirements.

Insulation resistance shall be not less than 1,000 megohms.

Capacitance to ground, and insertion loss (check test) shall be measured and shall meet the initial requirements.

Marking: Each filter shall be marked with the military part number, as shown in figure 2. In addition, full marking in accordance with method I requirements of MIL-STD-1285 shall be marked on each unit package.

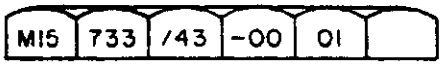


FIGURE 2. Hex flats-expanded view.

Part number: M15733/43- (dash number from table I).

QUALITY ASSURANCE PROVISIONS:

Qualification inspection: In accordance with MIL-F-15733.

Quality conformance inspection: In accordance with MIL-F-15733 except as follows:

Group A inspection for all dash numbers shall be in accordance with table II. Lots having greater than 10% rejects shall be considered reject lots. Reject lots, at the option of the manufacturer, may be reworked and submitted to group A inspection of table II.

TABLE II. Group A inspection.

Test or inspection	AQL (percent defective or 100% inspection)	
Voltage conditioning	100%	
Capacitance to ground	100%	
Dielectric withstanding voltage	100%	
Insulation resistance	100%	
Voltage drop	100%	
Insertion loss (check test)	100%	
	Major	Minor
Visual and mechanical inspection	1.0	4.0

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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